15-2-21

Lecture 1

Statistics and Data Description

Statistics:

Seligman defines statistics as the science which deals with the methods of collecting classifying, presenting, comparing and interpreting numerical data collected to throw some light on any sphere of enquiry.

The Merriam-Websters Collegiate Dictionary definition is: "A branch of mathematics dealing with the collection, analysis, interpretation and pre-sentation of masses of numerical data."

The American Heritage Dictionary defines statistics as: "The mathematics of the collection, organization and interpretation of and ~~presentation~~ of numerical data, especially the analysis of population characteristics by inference from sampling."

A simple but concise definition by Croxton and Cowden: "Statistics is de-fined as the collection,

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presentation, analysis and interpretation of numerical data."

A comprehensive definition of statistics can be collection. organisation. pre-sentation, analysis and interpretation of numerical data.

In other words it can be defined as-
" Statistics is the science and branch of applied mathematics which concerns with the collection of data, classifying and orgaization of data, presenting, comparing, finding the hidden information of data, interpreting and drawing interface of data.

Some important terms related to Statistics-

1) <u>Variable</u>- various values of the characteristic recorded from different unit are expressed by quantity, the characteristic is called variable.

- Qualitive variable
- Quantitive "
- Discrete variable
- Continous "

The variable, ~~@~~ Discrete variable or Continous variable may again classified as Random variable and Non-random variable.

2) Population: Total of units which are under investigation according to predetermind objective

3) Sample - Representative number of population units.

4) Data - Any measurment of one or more variable(s) recorded either from population units of sample units is known as data.

- Raw data - collected from sampling units and stored without any systematic fashion.
- Classified data - data are presented in a systematic fashion in rows or columns or even in ordered way.

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- Primary data - collected from sampling units by personal interview or mailed questionnaire.
- Secondary data - collected from official record or from published work.

⇒ Measurement and measurement scales of Data

- Nominal scale - The lowest measurement scale is nominal scale. As the name implies it consists of "naming" observations and classifing them into various mutally exclusive and collectively exhaustive catagories.
- Ordinal scale - Whenever observations are not of only different from category but category but can be ranked according to some ~~ei~~ criterion.
- Interval scale - The interval scale is more sophisticated scale than nominal or ordinal in that with this scale not only is it possible to orders measurments but also the distance between two measurements is known.

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- Ratio scale - The highest level of measurement is the ratio scale. This scale is characterized by ~~that~~ the fact that equality of ratios as well as equality of intervals may be determined. Fundamental to the ratio scale is a true zero point. The measurement of such familiar traits as heights, weights and length makes use of the ratio scale.

Let $x_1, x_2 \cdots x_n$ be the random sample from a population and $x_{(1)}, x_{(2)} \cdots x_{(n)}$ be the ordered sample. Then.

$$\text{Total} = x_1 + x_2 + \cdots + x_n = \sum_{i=1}^{n} x_i$$

$$\text{Product} = x_1 \times x_2 \cdots \times x_n = \prod_{i=1}^{n} x_i$$

$$\text{Range } R = \text{Maximum} - \text{Minimum}$$

$$= x_{(n)} - x_{(1)}$$

Sturge's Formula for number of class

$$k = 1 + 3.332 \log_{10}(n)$$

$$\text{class width, } h = \frac{R}{k}$$

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Arithmetic Mean, $AM = \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$

Geometric Mean, $GM = (x_1 x x_2 -- x x_n)^{\frac{1}{n}}$

$$= \left(\prod_{i=1}^{n} x_i\right)^{\frac{1}{n}}$$

$$GM = \text{Anti}\log \frac{1}{n} \sum_{i=1}^{n} \log_{10}(x_i)$$

$$= 10^{\left[\frac{1}{n}\sum_{n=1}^{i} \log_{10}(x_i)\right]}$$

Hermonic Mean, $HM = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}}$

$$= \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}$$

Trimmed Mean

Trimmed mean is compromise between $\bar{x}$ and $\tilde{x}$. A 10% trimmed mean, for example, would be computed by ~~elem~~ eliminating the smallest 10 percent and largest 10 percent of the sample and then averaging what is left over.

$\tilde{x} = \bar{x}_{tr(\alpha)} = \alpha$ ~~perce~~ percent trimmed mean.

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Winsorized Mean

Let $x_1, x_2 \cdots x_{10}$ be the random sample and $x_{(1)}, x_{(2)} \cdots x_{(10)}$ be ordered sample. Then the 10% winsorized mean is defined by -

$$WH = \frac{x_{(2)} + x_{(2)} + x_{(3)} + x_{(4)} + x_{(5)} + x_{(6)} + x_{(7)} + x_{(8)} + x_{(9)} + x_{(9)}}{10}$$

Tukey's Trimean or Trimean (TM)

$$TM = \frac{Q_1 + 2Q_2 + Q_3}{4}$$ where Q_i be the ith quartile.

Interquartile mean (IQM)

IQM or (midmean) is a statistical measure of central tendency based on the truncated mean of the interquartile range In calculation of the IQM, only the data between the first and third quartiles is used and the lowest 25% and highest 25% of data are discarded.

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let $x_1, x_2 - - x_n$ the random sample and $x_{(1)}, x_{(2)} - - x_{(n)}$ " orded " the

$$x_{IQM} = \frac{2}{n} \sum_{i=(\frac{n}{4}+1)}^{\frac{3n}{4}} x_{(i)}$$

Quartile

Quartile means the measure of location in the array of data set which divides the array into four equal parts. There are three quartile 1) First Q_1 2) second Q_2 3) Third Q_3.

if n is odd

$$Q_i = x_{(\frac{i(n+1)}{4})}$$

$$Q_i = \frac{1}{2}\left[x_{(\frac{in}{4})} + x_{(\frac{in}{4}+1)}\right]$$ if n is even

Inter quartile range, $IQR = Q_3 - Q_1$

Percentile

Percentile is the measure of the location in the array of data set which divides the array into hundred equal parts. There are 99 percentiles.

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$$P_i = x_{\left(\frac{i(n+1)}{100}\right)} \quad \text{if } n \text{ is odd}$$

$$P_i = \frac{1}{2}\left[x_{\left(\frac{in}{100}\right)} + x_{\left(\frac{in}{100}+1\right)}\right] \quad n \text{ is even}$$

Mode

Mode is the value of variable which occurs most frequently in series of observation of the variable.

Median

Median is the value of the ordered series which divides the array into two equal parts such that half of observation below it and other half of observation are ~~avot~~ above it.

$$Me = x_{\left(\frac{(n+1)}{2}\right)} \quad \text{if } n \text{ is odd}$$

$$Me = \frac{1}{2}\left[x_{\left(\frac{n}{2}\right)} + x_{\left(\frac{n}{2}+1\right)}\right] \quad n \text{ is even.}$$

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Ex 1: The data represents the weight (kg) of the 35 students of the dept of CSE. at JU: 36, 37, 40, 41, 41, 44, 45, 47, 48, 48, 50, 51, 51, 51, 51, 52, 55, 55, 55, 55, 57.5, 59, *60, *60, *64, *65, *67, *68, 70, 75, 75, 78, 78 and 87. Now include your weight (kg) with above data and evaluate - Mean, Median, Mode, Trimmed mean, winsorized mean, Interquartile mean, Quartiles, Inter Quartile Range, Trimean, P60

Solution:

After adding my weight (47) kg, the number of students are 36. And the list is -

36, 37, 40, 41, 41, 44, 44, 45, 47, 47, 48, 48, 50, 51, 51, 51, 51, 52, 55, 55, 55, 55, 57.5, 59, 60, 60, 64, 65, 67, 68, 70, 75, 75, 78, 78 and 87.

and $n = 36$

i) Mean -

$$\text{Mean} = \frac{\text{Total}}{n}$$

$$= \frac{\sum_{i=1}^{36} x_i}{n} = \frac{2007.5}{36}$$

$$= 55.76 \text{ kg}$$

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ii) Median-

$n = 36$ which is even,

So, $Me = \frac{1}{2}\left[x_{\frac{n}{2}} + x_{(\frac{n}{2}+1)}\right]$

$= \frac{1}{2}\left[x_{\frac{36}{2}} + x_{\frac{36}{2}+1}\right]$

$= \frac{1}{2}\left[x_{18} + x_{19}\right]$

$= \frac{1}{2}(52+55)$

$= 53.5$ kg

iii) Mode-

In the data of student weight list 51 and 55 are the most frequent series.

So, there are 2 mode which are 51 kg and 55 kg.

iv) Trimmed Mean [10 %] -

$n = 36$.

10% of $n = 36 \times \frac{10}{100} = 3.6 = 4$

Now, remove 4 sample from each side.

So, $n = 28$.

Now, 10% trimmed mean,

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$$\tilde{x} = \frac{41+44+44+45+47+47+48+48+50+51+51+51+51+51+52+55+55+55+55+57.5+59+60+60+64+65+67+68+70+75}{28}$$

$$= \frac{1535.5}{28}$$

$$\tilde{x} = 54.84 \text{ kg}.$$

v) Winsorized Mean (10%) WM -

$n = 36$. 10% of $n = 3.6 = 4$.

Now, replace first 4 sample with 5th sample and last 4 sample with 31st and evaluate mean

$$WM = \frac{41+41+41+41+41+44+44+45+47+47+48+48+50+51+51+51+51+52+55+55+55+55+57.5+59+60+60+64+65+67+68+70+75+75+75+75}{36}$$

$$\Rightarrow WM = \frac{1999.5}{36}$$

$$\therefore WM = 55.54 \text{ kg}$$

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vi) Interquartile Mean (IQM)-

here $n = 36$,

We know,

$$X_{IQM} = \frac{2}{n} \sum_{i=\frac{n}{4}+1}^{\frac{3n}{4}} x_i$$

$$= \frac{2}{36} \sum_{i=10}^{27} x_i$$

$$= \frac{2}{36} (47+48+48+50+51+51+51+51+52+55+55+55+55+57.5+59+60+60+64)$$

$$= \frac{2}{36} \times 969.5 \text{ kg}$$

$$X_{IQM} = 53.86 \text{ kg}.$$

vii) Quartiles -

here $n=36$ which is even.

So, $Q_i = \frac{1}{2}\left[x_{\frac{in}{4}} + x_{(\frac{in}{4}+1)}\right]$

$$Q_1 = \frac{1}{2}\left[x_{\frac{36}{4}} + x_{(\frac{36}{4}+1)}\right]$$

$$= \frac{1}{2}(x_9 + x_{10})$$

$$= \frac{1}{2}(47+47)$$

$$\therefore Q_1 = 47 \text{ kg}$$

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$$Q_2 = \frac{1}{2}\left[x_{\frac{2\times36}{4}} + x_{\left(\frac{2\times36}{4}+1\right)}\right]$$

$$= \frac{1}{2}\left[x_{18} + x_{19}\right]$$

$$= \frac{1}{2}(52+55)$$

$$\therefore \quad Q_2 = 53.5 \text{ kg}$$

$$Q_3 = \frac{1}{2}\left[x_{\frac{3\times36}{4}} + x_{\left(\frac{3\times36}{4}+1\right)}\right]$$

$$= \frac{1}{2}\left[x_{27} + x_{28}\right]$$

$$= \frac{1}{2}(64+65)$$

$$\therefore Q_3 = 64.5 \text{ kg}$$

viii) Inter Quartile Range (IQR) -

$$IQR = Q_3 - Q_1$$

$$= 64.5 \text{ kg} - 47 \text{ kg}$$

$$\therefore IQR = 17.5 \text{ kg}$$

ix) Trimean or Tukey's Trimean (TM)

$$TM = \frac{Q_1 + 2Q_2 + Q_3}{4}$$

$$= \frac{47 + 2\times 53.5 + 64.5}{4}$$

$$= \frac{218.5}{4}$$

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$\therefore$ TM = 54.625 kg

x) Percentile (P_{60})-

$n = 36$ which is even,

$\therefore$ $P_i = \frac{1}{2}\left[x_{\frac{in}{100}} + x_{\left(\frac{in}{100}+1\right)}\right]$

$\therefore P_{60} = \frac{1}{2}\left[x_{\frac{60\times36}{100}} + x_{\left(\frac{60\times36}{100}+1\right)}\right]$

$= \frac{1}{2}\left[x_{21.6} + x_2\right]$

$= \frac{1}{2}\left[x_{21.6} + x_{22.6}\right]$

$= \frac{1}{2}\left[x_{22} + x_{23}\right]$

$= \frac{1}{2}(55 + 57.5)$

$\therefore P_{60} = 55.25$ kg

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Frequency, distribution and Histogram

Ex: 2: The data represents the weight (kg) of the thirty five students: 36, 37, 40, 41, 41, 44, 44, 45, 47, 48, 48, 50, 51, 51, 51, 51, 52, 55, 55, 55, 55, 57.5, 59, 60, 60, 64, 65, 67, 68, 70, 75, 75, 78, 78 and 87. Using suitable class interval construct a frequency table. Also ~~constru~~ calculate relative frequency and cumulative frequency.

Solution: Here minimum = 36, maximum = 87

$$R = 87 - 36 = 51, \quad n = 35$$

number of class $k = 1 + 3.332 \log_{10}(n)$

$= 6.14 \simeq 7$

class width $h = \frac{R}{k}$

$= \frac{51}{7} = 8.30 \simeq 9$

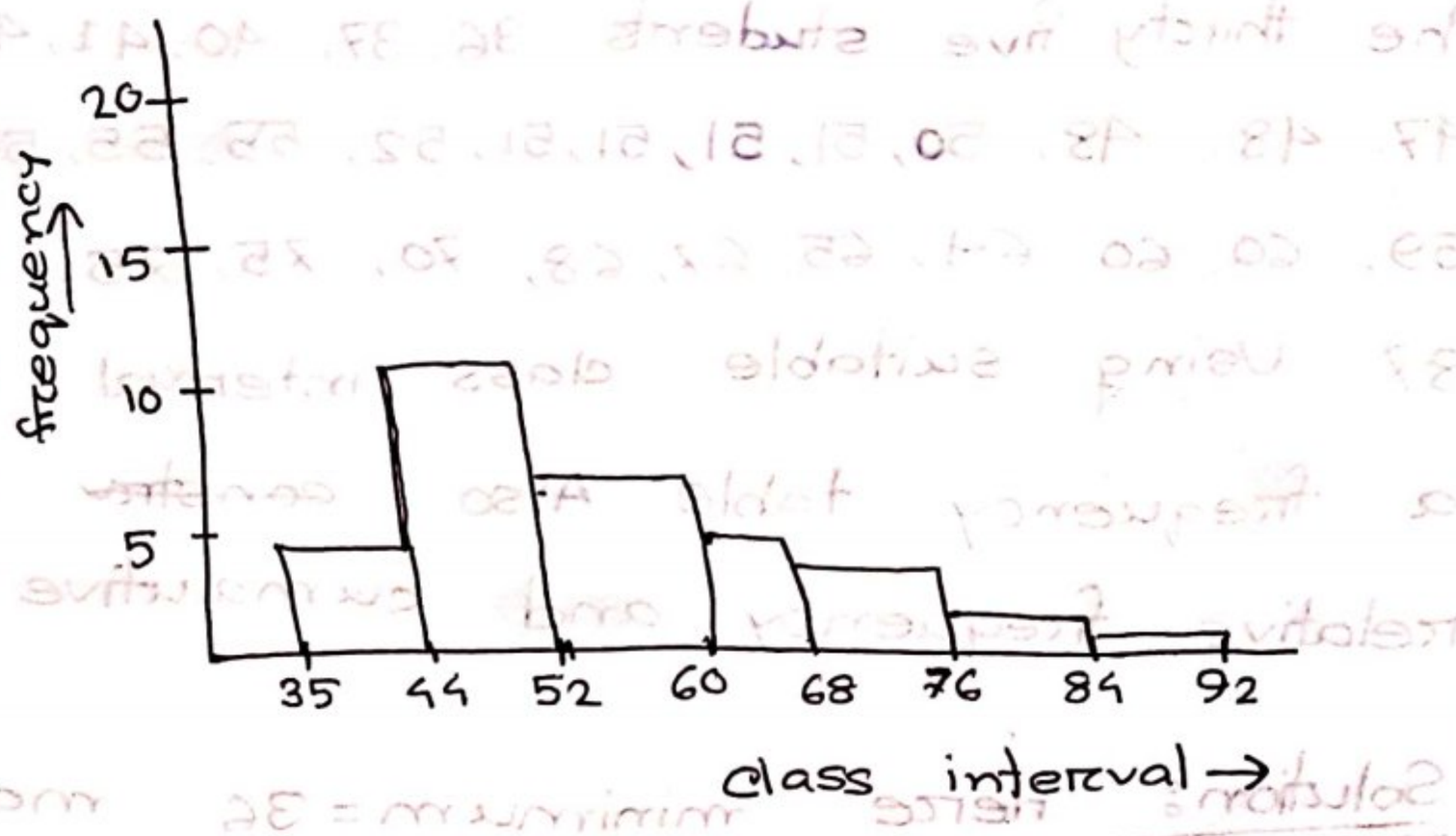
class interval	Frequency	Cumulative freq.	Relative freq. ($\frac{\text{freq}}{35}$)
35 - 44	5	5	0.143
44 - 52	11	16	0.314
52 - 60	7	23	0.200
60 - 68	5	28	0.143
68 - 76	4	32	0.114
76 - 84	2	34	0.057
84 - 92	1	35	0.029

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Histogram:



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Lecture-2

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Stem and leaf plot

Stem and leaf plots is a tabular presentation of data where each data value is split into a "stem" or "the first digit or digits" and a "leaf" or ~~usally~~ usually "the last digit". Procedure to make a stem and leaf plot is separate each digit a data into steam which will consist of all except the rightmost digit and leaf with the rightmost digit.

Example-

36, 37, 40, 41, 50, 87, 88, 110, 112

stem	leaf	freq
3	6, 7	2
4	0, 1	2
5	0	1
8	7, 8	2
11	0, 2	2

#Empirical relationship between Mean, Median, Mode presented by Karl Pearson-

$$Mode = 3Median - 2Mean$$

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The parceto Chart

The parceto diagram is a bare graph for count data. It displays the frequency of each count on the vertical axis and the category of classification on the horizontal axis. We always arrange the categories in desending order of freq. of occurrence, that is the most frequently occurring is on the left followed by the next most frequently occurring.

Ex: (Ex-2 Table)

class Interval	frequency	Mid	Percent	Cum %
49-52	11			
052-4560	7			

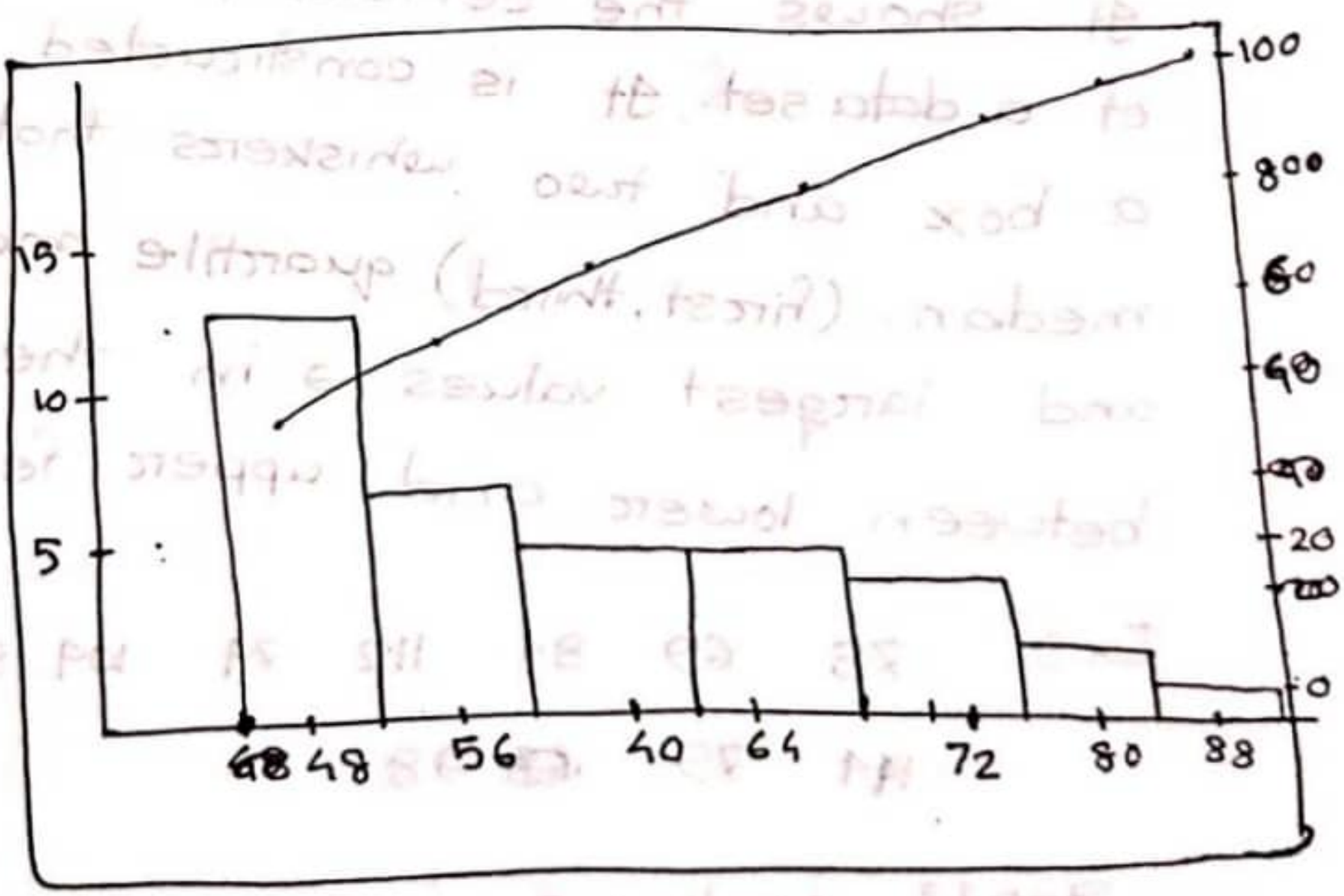
Note:

In diagram take middle point of class interval.

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$n = 36$

class Interval	Mid	Frq.	Percent	Cumulative percent
44-52	48	12	33.33	33.33
52-60	56	7	19.44	52.77
35-44	40	5	13.89	66.66
60-68	64	5	13.89	80.55
68-76	72	4	11.11	91.66
76-84	80	2	5.55	97.22
84-92	88	1	2.78	100.00



Box-and-Whisker Plot

A box and whisker plot gives a graphical presentation of data using five measures the median, the first quartile, the third quartile and smallest and largest values in the dataset between the lower and upper inner fences.

It shows the center, spread, and skewness of a data set. It is constructed by drawing a box and two whiskers that use the median, (first, third) quartile and smallest and largest values in the data set between lower and upper fences.

Ex: 75 69 84 112 79 109 81 90 94 144 79 98

Step1: Median =

69, 74, 75, 79, 81, 84, 90, 94, 98, 104, 112, 144

$$\text{Median} = \frac{1}{2}(84+90) = 87$$

$$Q_1 = (75+79)/2 = 77$$

$$Q_3 = (98+104)/2 = 101$$

$$IQR = Q_3 - Q_1 = 24$$

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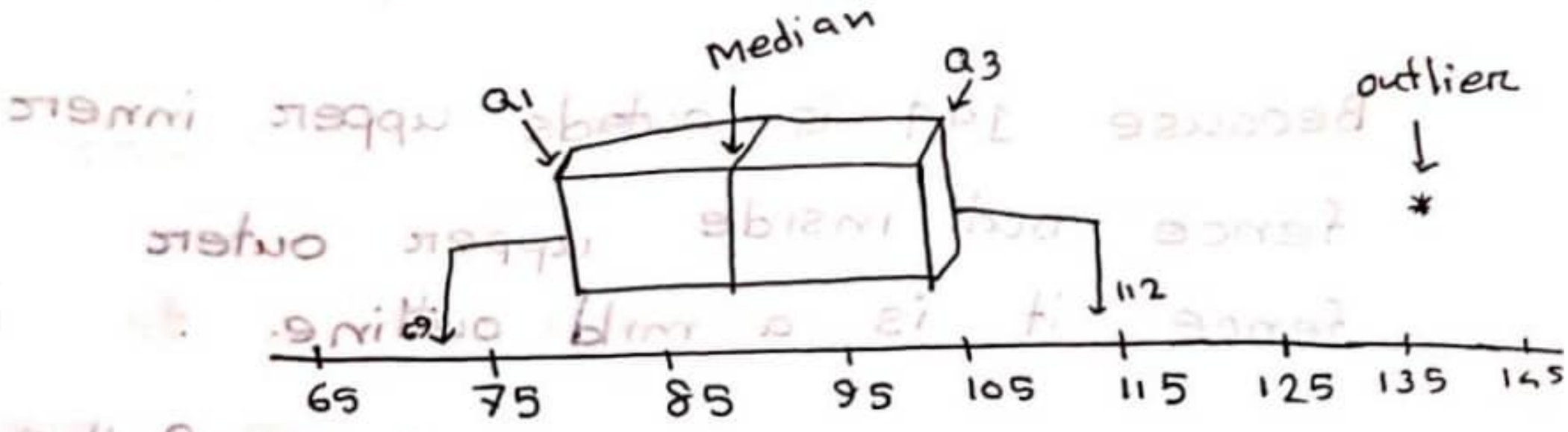
step 2: $1.5 \times IQR = 36$

Lower inner fence $= Q_1 - 36 = 91$

Upper " " $= Q_3 + 36 = 137$

step 3: Smallest value within two innerfence = 69

Largest " " " " " = 112



The observations that fall outside the two inner fences are called outlines. The outliners can be classified into two kind of outliers - mild and extreme outliers. To do so, we define two outer fences - a lower outer fence at $3.0 \times IQR$ below the first quartile and upper outer fence at $3 \times IQR$ above the third quartile.

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* If an observation is outside ~~eight~~ either of the two inner fences but within either two outer fences, it is called a mild outlier.

Ex: for above example the outer fences are → ~~mild outliers~~ {75 - (3×24)} = 5 and

{101+(3×24)} = 173

Because 199 is outside upper inner fence but inside upper outer fence, it is a mild outline.

* An observation is outside either of the two outer fences is called extreme outlier.

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8-4.2 Measures of Dispersion

sample variance

$$S^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1} = \frac{S_{xx}}{n-1}$$

Variance:

Variance measures the average amount of variation of observations. standard diviation measures the average distance of the observations from mean.

if S^2 is small, then there is relatively little variability in the data. if s^2 is large, the variability is relatively large.

$$S_{xx} = SS(x) = \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$$

$$S_{XY} = SP(xy) = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}$$

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Example: The data represents the weight (kg) of the 36 students. of dept of CSE : 36, 37, 40. 41, 41, 44, 44, 45, 47, 47, 48, 48, 50, 51, 51, 51, 51, 52, 55, 55, 55, 55, 58, 59, 60, 60. 64, 65. 67, 68, 70, 75, 75, 78, 78 and 87.

Now calculate Stem and leaf plot, Histogram, Pareto chart. Box and whiskers plot, mild and extreme outliers. standard deviation.

Solution:

a) stem and leaf plot:

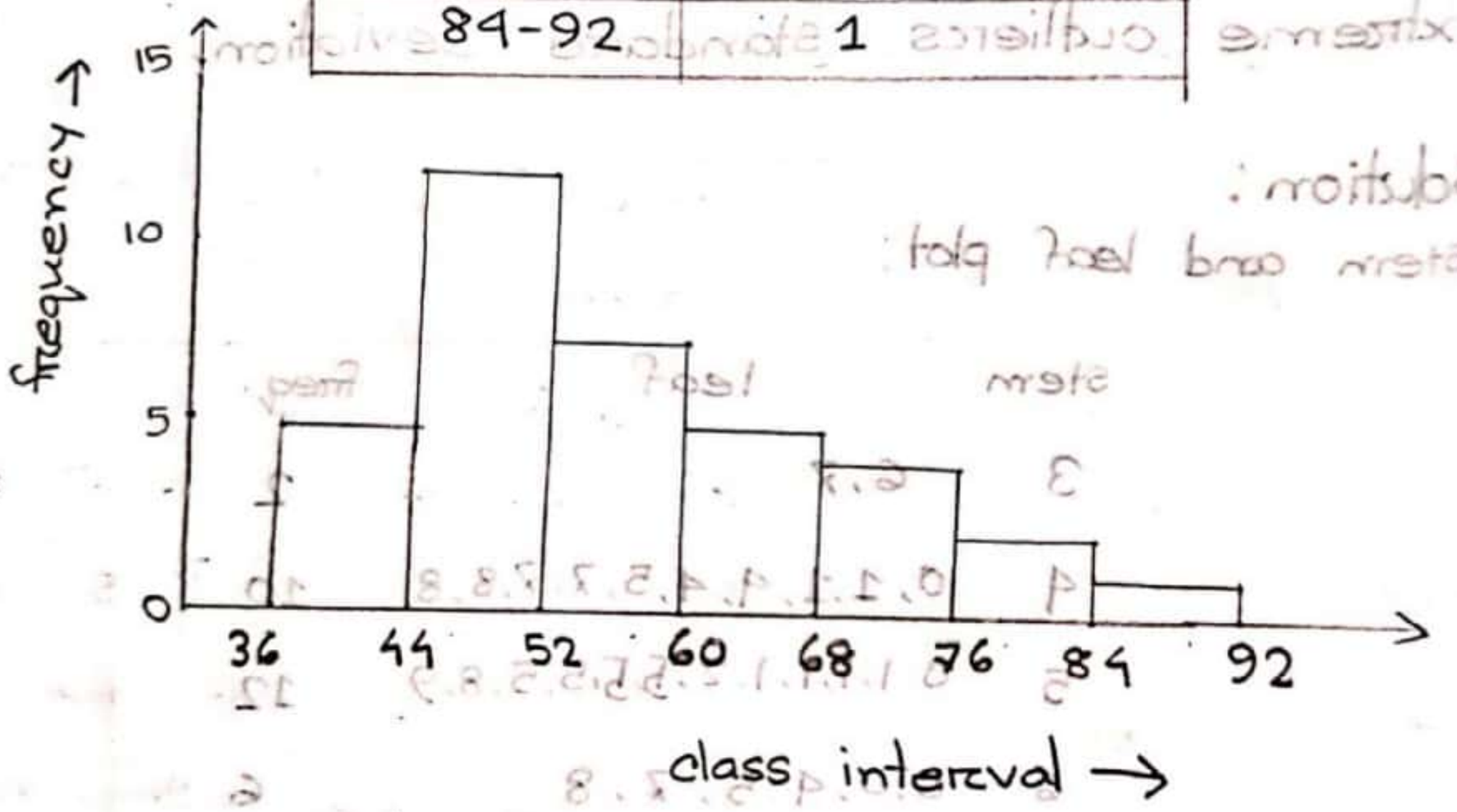
stem	leaf	freq.
3	6, 7	2
4	0, 1, 1, 4, 4, 5, 7, 7, 8, 8	10
5	0, 1, 1, 1, 1, 2, 5, 5, 5, 5, 8, 9	12
6	0, 0, 4, 5, 7, 8	6
7	0, 5, 5, 8, 8	5
8	7	1

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Histogram:

Class Interval	Frequency
36-44	5
44-52	12
52-60	7
60-68	5
68-76	4
76-84	2
84-92	1

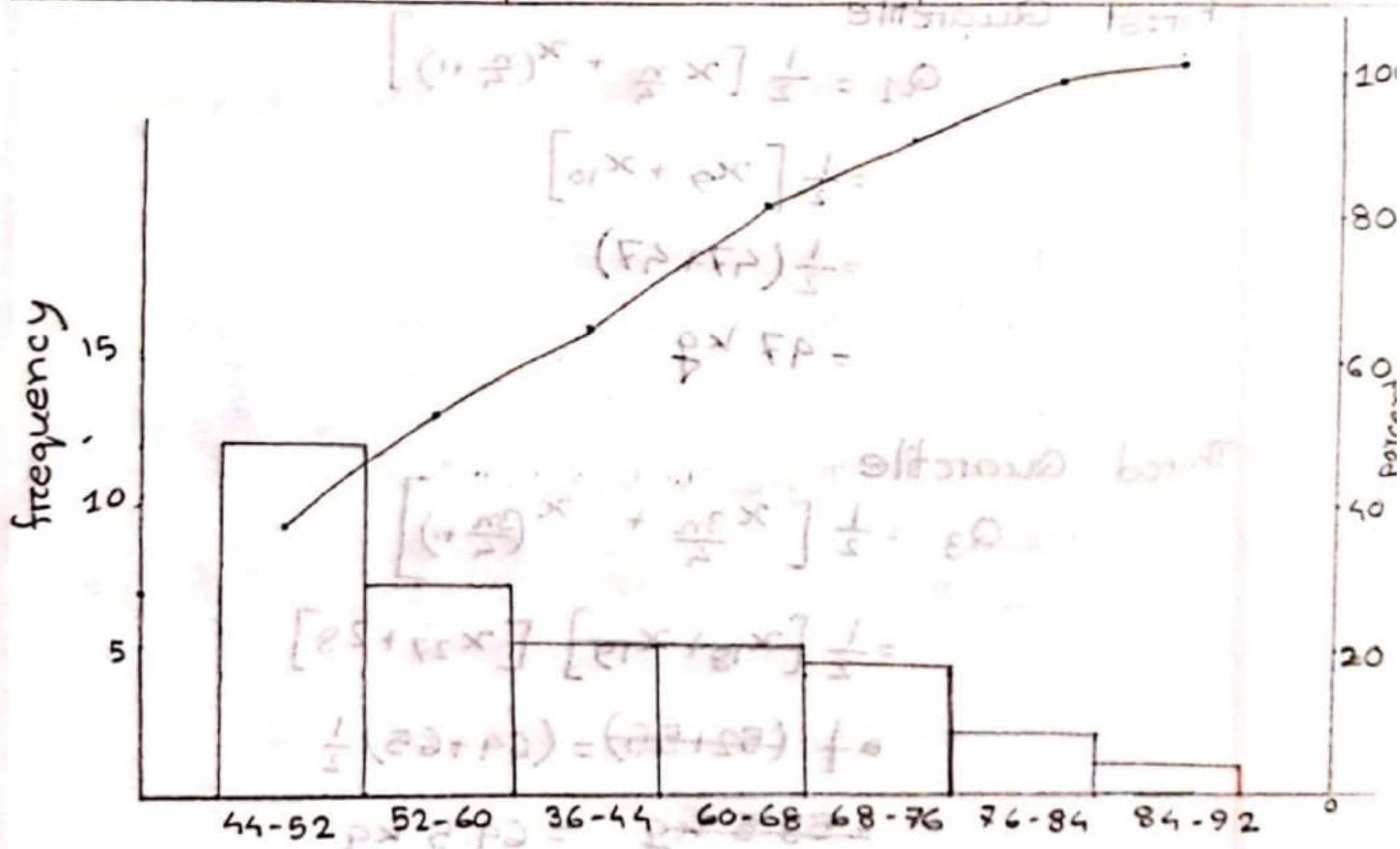


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#Pareto chart

class interval	Frequency	Parcent (%)	Cumulative parcen (%)
44-52	12	33.33	33.33
52-60	7	19.44	52.78
36-44	5	13.89	66.67
60-68	5	13.89	80.56
68-76	4	11.11	91.67
76-84	2	5.56	97.22
84-92	1	2.78	100



Parcent : 33.33 19.44 13.89 13.89 11.11 5.56 2.78

Cumulative %: 33.33 52.78 66.67 80.56 91.67 97.22 100

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Box and whisker plot, mild and extreme outliers

Step 1:

here n is 36 which is even.

$\therefore$ Median $= \frac{1}{2}\left[x_{\frac{n}{2}} + x_{(\frac{n}{2}+1)}\right]$

$= \frac{1}{2}\left[x_{18} + x_{19}\right]$

$= \frac{1}{2}(52+55) = 53.5$ kg

First Quartile

$Q_1 = \frac{1}{2}\left[x_{\frac{n}{4}} + x_{(\frac{n}{4}+1)}\right]$

$= \frac{1}{2}\left[x_9 + x_{10}\right]$

$= \frac{1}{2}(47+47)$

$= 47$ kg

Third Quartile

$Q_3 = \frac{1}{2}\left[x_{\frac{3n}{4}} + x_{(\frac{3n}{4}+1)}\right]$

$= \frac{1}{2}\left[x_{27}+28\right]$

$= \frac{1}{2}(52+55) = (64+65)\frac{1}{2}$

$= 53.5$ kg $= 64.5$ kg

$IQR = Q_3 - Q_1$

$= (64.5 - 47)$ kg

$= 17.5$ kg

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Step 2:

Lower inner fence $= Q_1 - (1.5 \times IQR)$

$= 47 - 26.25$

$= 20.75$

upper " " $= Q_3 + (1.5 \times IQR)$

$= 64.5 + 26.25$

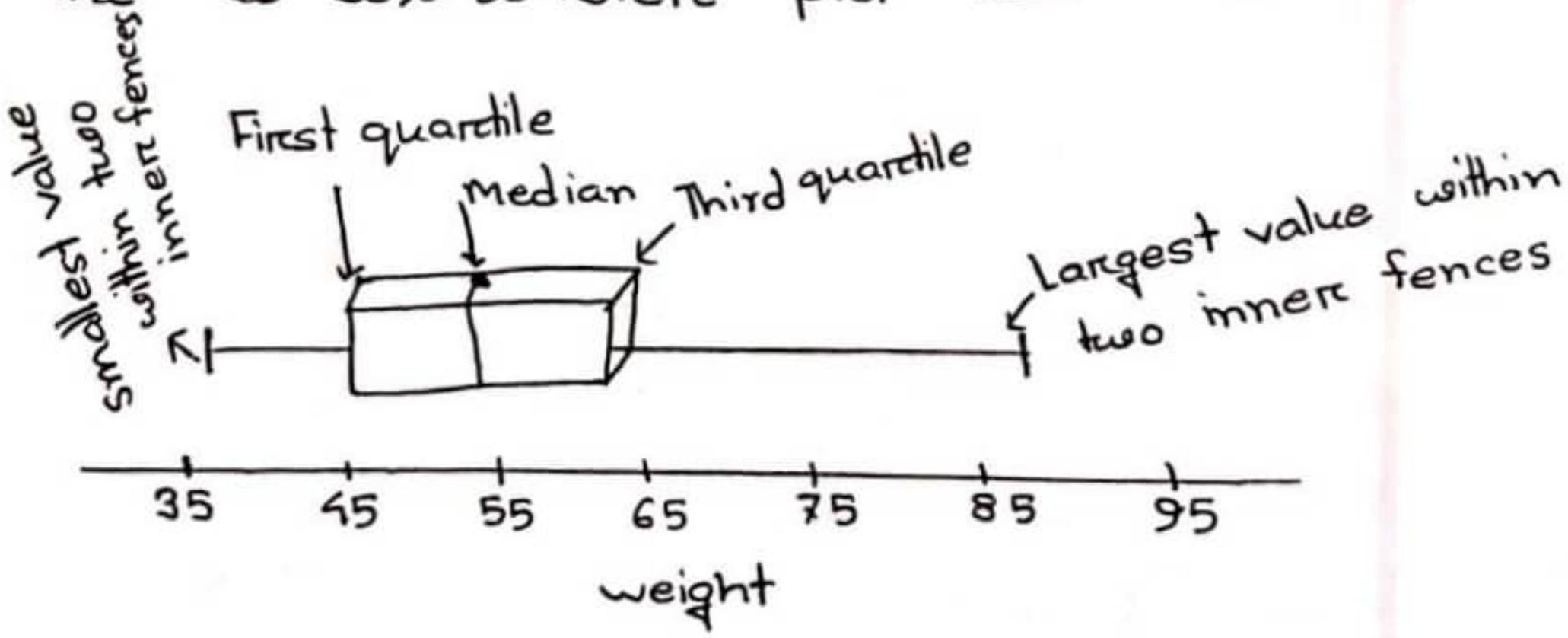
$= 90.75$

Step 3:

smallest value within two inner fences $= 36$

Largest " " " " " $= 87$

So, the box-whisker plot will be -



All observations are within two inner fences.

So, there are no outliers.

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Sample standard div deviation :

b) we know.

⇒ <u>sample variance</u>.

$$s^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}$$

⇒ <u>Sample Mean :</u>

$$\bar{x} = \frac{\sum_{i=1}^{n=36} x_i}{36} = \frac{2008.5}{36} = 55.78 \text{ kg}$$

Weight, x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
36	-19.78	391.16
37	-18.78	352.27
40	-15.78	249.0
41	-14.78	218.44
41	-14.78	218.44
44	-11.78	138.77
44	-11.78	138.77
45	-10.78	116.21
47	-8.78	77.09
47	-8.78	77.09
48	-7.78	60.53
48	-7.78	60.53
50	-5.78	33.41
51	-4.78	22.85
51	-4.78	22.85
51	-4.78	22.85
51	-4.78	22.85
52	-3.78	14.28

x_i	$(x_i - \bar{x})$	$(x_i - \bar{x})^2$
55	-0.78	0.609
55	-0.78	0.609
55	-0.78	0.609
55	-0.78	0.609
58	2.22	4.93
59	3.22	10.37
60	4.22	17.81
60	4.22	17.81
64	8.22	67.57
65	9.22	85.00
67	11.22	125.89
68	12.22	149.33
70	14.22	202.21
75	19.22	369.41
75	19.22	369.41
78	22.22	493.73
78	22.22	987.46
87	31.22	974.69

Sum = 5607.85

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Now, $s^2 = \frac{\sum_{i=1}^{n=36} (x_i - \bar{x})^2}{36-1}$

$= \frac{5607.85}{35}$

$= 160.22$

So,
sample standard deviation,

$s = \sqrt{s^2}$

$= \sqrt{160.22}$

$= 12.66$ kg.

c) Ordered stem and leaf plot-

stem	leaf
3	6, 7
4	0, 1, 1, 4, 4, 5, 7, 7, 8, 8
5	0, 1, 1, 1, 1, 2, 5, 5, 5, 5, 8, 9
6	0, 0, 4, 5, 7, 8
7	0, 5, 5, 88
8	7

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8-3-21

Lecture-3

page -185 পড়বা.

Population Mean. $\mu = \frac{1}{N} \sum_{i=1}^{N} x_i$

Population Variance: $\sigma_x^2 = \frac{1}{N} \sum_{i=1}^{N} (x_i - \mu)^2$

Population standard dieviation $\sigma_x = \sqrt{\frac{1}{N} \sum_{x=1}^{N} (x_i - \mu)^2}$

Population Covariance $cov(xy) = \frac{1}{N} \sum_{i=1}^{N} (x_i - \mu)(y_i - \mu)$

Sample mean: $\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$

Sample variance. $s_x^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$

Sample Standard ~~Dive~~ Deviation: $s_x = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2}$

Sample covariance: $cov(xy) = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$

Measure of Dispersion

The term dispersion means the scatteredness of observations from some central value. The term "measure of dispension" is defined as an average of deviation of observations from some central value. There are two types

1) Absdute measure of dispersion
2) Relative measure of dispersion.

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Absolute measure of dispersion

- Range $R = x_{(n)} - x_{(1)}$
- Mean deviation

$$MD\ (Mean) = \frac{1}{n}\sum_{i=1}^{n} |x_i - \bar{x}|$$

$$MD\ (Median) = \frac{1}{n}\sum_{i=1}^{n} |x_i - Me|$$

$$MD\ (Mode) = \frac{1}{n}\sum_{i=1}^{n} |x_i - Mo|$$

- Standard deviation, $s_x = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n=1} (x_i - \bar{x})^2}$
- Quartile deviation $QD = \frac{Q_3 - Q_1}{2}$

#

Relative measure of dispersion

- Coefficient of Range $= \frac{x_{(n)} - x_{(1)}}{x_{(n)} + x_{(1)}} \times 100\ \%$

Coefficient of mean Deviation

coefficient of $MD(Mean) = \frac{MD(mean)}{\bar{x}} \times 100\%$

coefficient of $MD(Median) = \frac{MD(Median)}{Median} \times 100\ \%$

coefficient of $MD(mode) = \frac{MD(mode)}{Mode} \times 100\ \%$

- Coefficient of variation. $CV = \frac{s_x}{\bar{x}} \times 100\%$
- Coefficient of Quartile Deviation, $\frac{Q_3 - Q_1}{Q_3 + Q_1} \times 100\%$

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$$\mu_1' = \bar{x}$$
$$\mu_2 = \frac{1}{N} \sum x_i^2$$

Coefficient of variation

A measure of relative variation called sample coefficient of variation. It is defined as-

$$CV = \frac{Sx}{\bar{x}} \times 100\%$$

Shape characteristics of Distribution

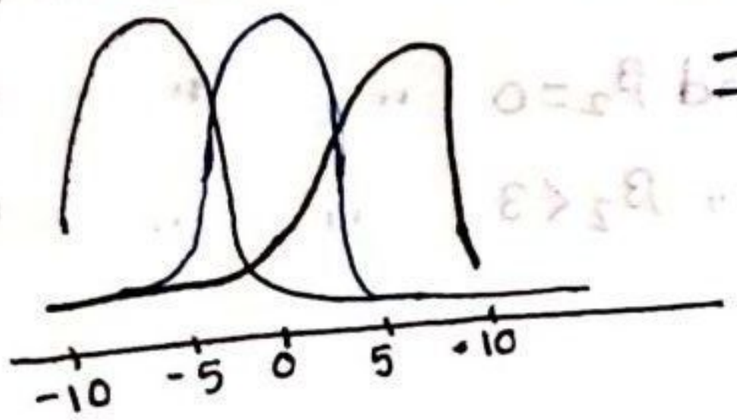
The important measure of dispersion is variance which is known as mean square ~~div~~ deviation or mean deviation of second order. The mean deviation of r-th order can be calculated and which is known as r-th central moment.

$$\mu_r = \frac{1}{N} \sum_{i=1}^{N} (x_i - \mu)^r$$
$$\mu_r' = \frac{1}{N} \sum x_i^r \quad [\text{raw moment}]$$

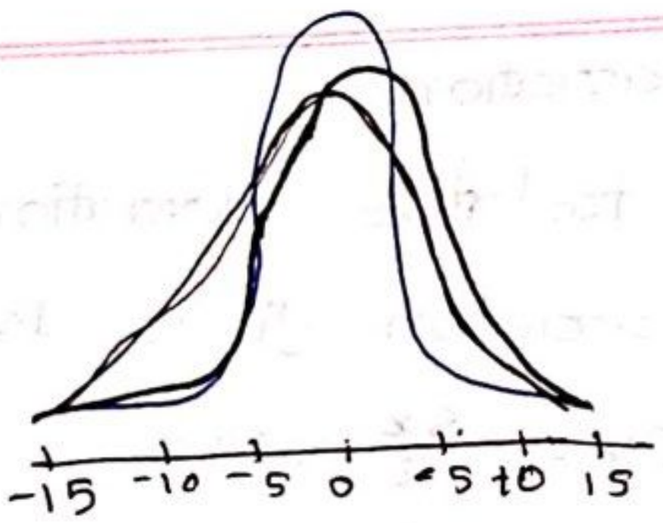
#Skewness and kurtosis

The asymmetric property of the curve is known as skewness and height characteristic of frequency curve is known kurtosis

- positive skewed
= symmetic
= Negative "



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— Lepto kurtise
— Meso "
— platy "

figure: kurtosis

~~Karl~~ perc

karl pearson's coefficient of skewness,

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} \text{ and } \gamma_1 = \sqrt{\beta_1} = \frac{\mu_3}{\sigma^3}$$

if $\gamma_1 < 0$ and $\mu_3 < 0$ then the distribution is negatively skewed

if $\gamma_1 = 0$ and $\mu_3 = 0$ then distribution is symmetric

if $\gamma_1 > 0$ and $\mu_3 > 0$ " " is positively skewed

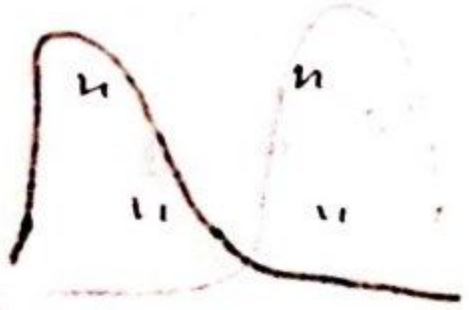
karl pearson's coefficient of kurtosis,

$$\beta_2 = \frac{\mu_4}{\mu_2^2} \text{ and } \gamma_2 = \beta_2 - 3 = \frac{\mu_4}{\sigma^4} - 3$$

if $\gamma_2 > 0$ and $\beta_2 > 3$ then distribution is Lepto kurtic

if $\gamma_2 = 0$ and $\beta_2 = 0$ " " " Meso "

if $\gamma_2 < 0$ " $\beta_2 < 3$ " " " Platy "



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If the data represent measurements on all the units of a finite population or a census, then these equations of karl pearson's coeifficient of skewness and kurtosis should be utilized directly to determine these measure

If the data being analyzed represent measurement of a variable made a sample units, the sample estimate of γ_1 and γ_2 are -

Skewness $\gamma_1 = \frac{n}{(n-1)(n-2)} * \frac{\sum_{i=1}^{n} (x_i - \bar{x})^3}{s_x^3}$ for $n>2$

kurtosis, $\gamma_2 = \frac{n(n+1)}{(n-1)(n-2)(n-3)} * \frac{\sum_{i=1}^{n} (x_i - \bar{x})^4}{s_x^4} - 3$ for $n>3$

these values are calculated from Excel.

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Tamashil
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Q 8.12: The data represents the weight (kg) of the students including my weight (47 kg) of the students of CSE dept of 30 -

36, 37, 90, 41, 41, 44, 44, 45, 47, 47, 48, 48, 50, 51, 51, 51, 51, 52, 55, 55, 55, 55, 58, 59, 60, 60, 64, 65, 67, 68, 70, 75, 75, 78, 78 and 87.

Now, calculate these -

a) ~~Stea~~ Stem and leaf plot -

Stem	leaf	frequency
3	6, 7	2
4	0, 1, 1, 4, 4, 5, 7, 7, 8, 8	10
5	0, 1, 1, 1, 1, 2, 5, 5, 5, 5, 8, 9	12
6	0, 0, 4, 5, 7, 8	6
7	0, 5, 5, 8, 8	5
8	7	1
		36

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b) Sample mean:

$$\text{Mean } \bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

$$= \frac{1}{36} \sum_{i=1}^{36} x_i$$

$$= \frac{2008}{36}$$

$$= 55.78 \text{ kg}$$

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d) Median and first and third quartile:

$n = 36$ which is even.

So, Median, $Me = \frac{1}{2}\left[x_{\frac{n}{2}} + x_{(\frac{n}{2}+1)}\right]$

$= \frac{1}{2}\left(x_{\frac{36}{2}} + x_{(\frac{36}{2}+1)}\right]$

$= \frac{1}{2}\left(x_{18} + x_{19}\right]$

$= \frac{1}{2}(52+55)$

$= 53.5 \text{ kg}$

first Quartile, $Q_1 = \frac{1}{2}\left[x_{\frac{n}{4}} + x_{(\frac{n}{4}+1)}\right]$

$= \frac{1}{2}\left(x_9 + x_{10}\right)$

$= \frac{1}{2}(47+47)$

$= 47 \text{ kg}$

Third Quartile, $Q_3 = \frac{1}{2}\left[x_{\frac{3n}{4}} + x_{(\frac{3n}{4}+1)}\right]$

$= \frac{1}{2}\left(x_{27} + x_{28}\right)$

$= \frac{1}{2}(69+65)$

$= 69.5 \text{ kg}$

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e) Interquartile range,

$$IQR = Q_3 - Q_1$$
$$= (64.5 - 47)\ kg$$
$$= 17.5\ kg$$

f) Skewness and kurtosis measures

here $n = 36$

$$\text{Skewness, } \gamma_1 = \frac{n}{(n-1)(n-2)} \frac{\sum_{i=1}^{n} (x_i - \bar{x})^3}{Sx^3}$$

$$\text{kurtosis, } \gamma_2 = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \frac{\sum_{i=1}^{n} (x_i - \bar{x})^4}{Sx^4} - 3$$

Now, $\bar{x} = 55.78\ kg$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^3$	$(x_i - \bar{x})^4$
36	-19.78	-7738.9	153075.3
37	-18.78	-6623.5	124389.12
40	-15.78	-3929.4	62005.18
41	-14.78	-3228.7	47719.70
44	-11.78	-1634.7	19256.7
44	-11.78	-1634.7	19256.7
45	-10.78	-1252.7	13504.39
47	-8.78	-676.8	5942.6
47	-8.78	-678.8	5942.6
48	-7.78	-470.9	3663.68
48	-7.78	-470.9	3663.68

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x_i	$x_i - \bar{x}$	$(x_i - x)^3$	$(x_i - \bar{x})^4$
50	-5.78	-193.1	1116.12
51	-4.78	-109.2	522.04
51	-4.78	-109.2	522.04
51	-4.78	-109.2	522.09
51	-4.78	-109.2	522.09
52	-3.78	-54.01	204.16
55	-0.78	-.47	0.37
55	-0.78	-.47	0.37
55	-0.78	-.47	0.37
55	-0.78	-.47	0.37
55	-0.78	-.47	0.37
58	2.22	10.9	24.28
59	3.22	33.39	107.50
60	4.22	75.15	317.14
60	4.22	75.15	317.14
64	8.22	555.4	4565.5
65	9.22	783.78	7226.42
67	11.22	1412.5	15847.89
68	12.22	1824.79	22298.9
70	14.22	2875.40	40888.23
75	19.22	7100.03	136462.5
75	19.22	7100.03	136462.5
78	22.22	10970.6	10970.6
78	22.22	10970.6	10970.6
87	31.22	30429.78	950017.47
	sum $\lvert x_i - \bar{x} \rvert$ = 366.22	42002.54	2311993.66

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Now, $S = 12.66$ kg

So, Skewness $\gamma_1 = \frac{n \sum_{i=1}^{n} (x_i - \bar{x})^3}{(n-1)(n-2) \; s^3}$

$= \frac{36 \times 42002.54}{35 \times 34 \times 12.66^3}$

$= 0.6262$

kurtosis, $\gamma_2 = \frac{n(n+1) \sum_{i=1}^{n} (x_i - \bar{x})^4}{(n-1)(n-2)(n-3) \; s^4} - 3$

$= \frac{36 \times 37 \times 2311993.66}{35 \times 34 \times 33 \times (12.66)^4} - 3$

$= 3.0528 - 3$

$= 0.0528$

4) Coefficient of variation

$$CV = \frac{Sx}{\bar{x}} \times 100\%$$

here, Standard deviation, $Sx = 12.66$

Sample mean, $\bar{x} = 55.78$

$\therefore CV = \frac{12.66}{55.78} \times 100\%$

$= 22.7\%$

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Mean Deviation :

$$MD(\text{mean}) = \frac{1}{n} \sum_{i=1}^{n} |x_i - \bar{x}|$$

$$= \frac{1}{36} \times 366.22$$

$$= 10.17 \text{ kg}$$

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15-3-21

"Lecture - 4"

Grouped Data : Frequency table

class interval	Mid, x_i	Frequency, f_i	Cumulative frequency
$x - [x+1h]$	x_1	f_1	f_1
$[x+1h] - [x+2h]$	x_2	f_2	f_1+f_2
$[x+2h] - [x+3h]$	x_3	f_3	$f_1+f_2+f_3$
⋮			
$[x+(k-1)h] - [x+kh]$	x_k	f_k	$\sum_{i=1}^{k} f_i$

total $\sum_{i=1}^{k} f_i = n$

Mean, Median for grouped data

Arithmetic Mean, $AM = \bar{x} = \frac{1}{n} \sum_{i=1}^{k} f_i x_i$

Geometric Mean, $GM = (x_1^{f_1} x_2^{f_2} \dots x_k^{f_k})$

$= 10^{\left[\frac{1}{n} \sum_{i=1}^{k} f_i \log(x_i)\right]}$

Harmonic Mean, $HM = \frac{n}{\sum_{i=1}^{k} \frac{f_i}{x_i}}$

⇒ Median, $M_e = L + \frac{h}{f}\left[\frac{n}{2} - c\right]$

Median is the value falling in a class for which cumulative frequency is just greater than or equal to $\frac{n}{2}$. Here L = lower limit of median class. h = width of median class. f = frequency of median class. c =

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corn cumulative of the class preceding of the median class.. n = total frequency

Mode for group data

$$\text{Mode, } Mo = L + \frac{h(f_m - f_p)}{(f_m - f_p) + (f_m - f_a)}$$

the class which having the maximum freq. in a frequency distribution is known as Modal class.

L = lowere limit of modal class

h = width of modal class

f_p = frequency of the class preceding the m Cla

f_m = n of modal class

f_a = n of the class following modal class

Quartile of grouped data

$$\text{Quartile, } Q_i = L + \frac{h}{f}\left[\frac{in}{4} - c\right], \quad i = 1, 2, 3.$$

Q_i class in that class for which cumulative frequency is just greater than or equal t $\frac{ni}{4}$. Here L = Lowere limit of Q_i class h = width of the Q_i class. f = frequency of Q_i class. c = cumulative of the class preceding of th Q_i class n = total frequency.

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Variance, Standard deviation, central moment for grouped data

Sample Variance $Var(x) = s^2 = \frac{1}{n-1} \sum_{i=1}^{k} f_i (x_i - \bar{x})^2$

$$s^2 = \frac{1}{n-1}\left[\sum_{i=1}^{k} f_i x_i^2 - \frac{\left(\sum_{i=1}^{k} f_i x_i\right)^2}{n} \right]$$

Sample variance deviation, $s = \sqrt{Var(x)} = \sqrt{s^2}$

Central moment $\mu_r = \frac{1}{N} \sum_{i=1}^{k} f_i (x_i - \bar{x})^r$, $r = 0, 1, 2, 3, 4$

$\mu_0 = 1$, $\mu_1 = 0$, $\mu_2 = \sigma^2$

Example -

minimum = 36, maximum = 87,

$R = 87 - 36 = 51$, $n = 35$,

$k = 1 + 3.332 \log_{10}(n) = 7$ $\quad h = \frac{R}{k} = \frac{51}{7} = 9$

cl	x_i	f_i	cf	$f_i x_i$	$f_i (x_i - \bar{x})^2$	$f_i (x_i - \bar{x})^3$	$f_i (x_i - \bar{x})^4$
35-44	39.5	5	5	197.5	1428.1	-24134	407865.4
44-52	48	11	16	528.0	776.2	-6619.7	57765.8
52-60	56	7	23	392.0	1.1	- 0.4	0.2
60-68	64	5	28	320.0	288.8	2194.9	16681.1
68-76	72	4	32	288.0	973.4	15185.7	236896.4
76-84	80	2	34	160.0	1113.9	26288.5	620408.9
84-92	88	1	35	88.0	998.6	31554.5	997122.1
Total		35		1973.5	5580.1	44569.5	2333739.9

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Mean $= \bar{x} = \frac{1}{n} \sum_{i=1}^{k} f_i x_i = \frac{1}{35} 1973.5 = 56.4$

Median. $Me = L + \frac{h}{f}\left[\frac{n}{2} - c\right]$

Median class 52-60

$L = 52,\ h = 9,\ f = 7,\ c = 16,\ \frac{n}{2} = 35$

$\therefore Me = 52 + \frac{9}{7}\left[\frac{35}{2} - 16\right]$

$= 52 + 1.93$

$= 53.9$

Mode $Mo = L + \frac{h(fm - fp)}{(fm - fp) + (fm - fa)}$

Modal class 44-52

$L = 44$ $\quad fa = 7$

$h = 9$

$fm = 11$

$fp = 5$

$Mo = 44 + \frac{9(11 - 5)}{(11 - 7) + (11 - 5)}$

$= 44 + 5.4 = 49.4$

Since Mean > Median > Mode. So, distribution is positively skewed.

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Third Quartile

$$Q_3 = L + \frac{h}{f}\left(\frac{3n}{4} - c\right)$$

Third Quartile class : 60-68

$L = 60 \quad f = 5$

$h = 9 \quad c = 23$

$$Q_3 = 60 + \frac{9}{5}(26.25 - 23)$$

$$= 65.85$$

Standard deviation.

Sample variance, $Var(x) = s^2 = \frac{1}{n-1}\sum_{i=1}^{k} f_i(x_i - \bar{x})^2$

$$= \frac{5580.1}{35-1} = 164.1$$

" standard deviation

$$s = \sqrt{Var(x)} = 12.8$$

Skewness and kurtosis

Let assume the sample mean is unbiased estimate of population mean.

$$\mu_2 = \frac{1}{35}\sum_{i=1}^{k=7} f_i(x_i - \bar{x})^2 = 164.1$$

$$\mu_3 = \frac{1}{35}\sum_{i=1}^{k=7} f_i(x_i - \bar{x})^3 = 1273.9$$

$$\mu_4 = \frac{1}{35}\sum_{i=1}^{k=7} f_i(x_i - \bar{x})^4 = 66678.2$$

ution

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Skewness $\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{1273.4^2}{164.1^3} = 0.37$

$\gamma_1 = \sqrt{\beta_1} = \frac{\mu_3}{\sigma^3} = 0.61 > 0$

kurtosis $\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{66678.2}{(164.1)^2} = 2.5$

$\gamma_2 = \beta_2 - 3 = 2.5 - 3 = -0.5 < 0$

since $\gamma_1 > 0$ and $\gamma_2 < 0$ so the distribution is positively skewed and platy kurtic.

Tama shil

ID: 392

8.10) The samples after sorting-

83.4, 84.3, 85.3, 86.7, 86.7, 87.4, 87.5, 87.6, 87.7
87.8, 87.9, 88.2, 88.3, 88.3, 88.3, 88.4, 88.5
88.5, 88.6, 88.6, 88.7, 88.9, 89.0, 89.2,
89.3, 89.3, 89.6, 89.7, 89.8, 89.8, 89.9, 89.9,
90.0, 90.1, 90.1, 90.3, 90.3, 90.4, 90.4, 90.4,
90.5, 90.6, 90.7, 90.8, 90.9, 91.0, 91.0, 91.0,
91.1, 91.1, 91.1, 91.2, 91.2, 91.5, 91.6, 91.6, 91.8, 91.8,
92.2, 92.2, 92.2, 92.3, 92.6, 92.7, 92.7, 92.7,
93.0, 93.2, 93.3, 93.3, 93.4, 93.7, 94.2,
94.2, 94.4, 94.7, 95.6, 96.1, 98.8, 100.3

a) Stem-and-leaf plot

Stem		leaf
83	—	4
84	—	3
85	—	3
86	—	7,7
87	—	4,5,6,7,8,9
88	—	2,3,3,3,4,5,5,6,6,7,9
89	—	0,2,3,3,6,7,8,8,9,9

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stem		leaf
90	——	0,1,1,3,3,4,4,4,5,6,7,8,9
91	——	0,0,0,1,1,1,2,2,5,6,6,8,8
92	——	2,2,2,3,6,7,7,7
93	——	0,2,3,3,4,7
94	——	2,2,4,7
95	——	6
96	——	1
98	——	8
100	——	3

b) frequency distribution and histogram

minimum number = 83.4

maximum = 100.3

$n = 80$

Range, $R = 100.3 - 83.4$

$= 16.9$

$k = 1 + 3.332 \log_{10} n$

$= 1 + 3.332 \log_{10} 80$

$= 7.34 \approx 8$

$\therefore$ width of class, $h = \frac{R}{k} = \frac{16.9}{7.34}$

≈ 2.3

≈ 3

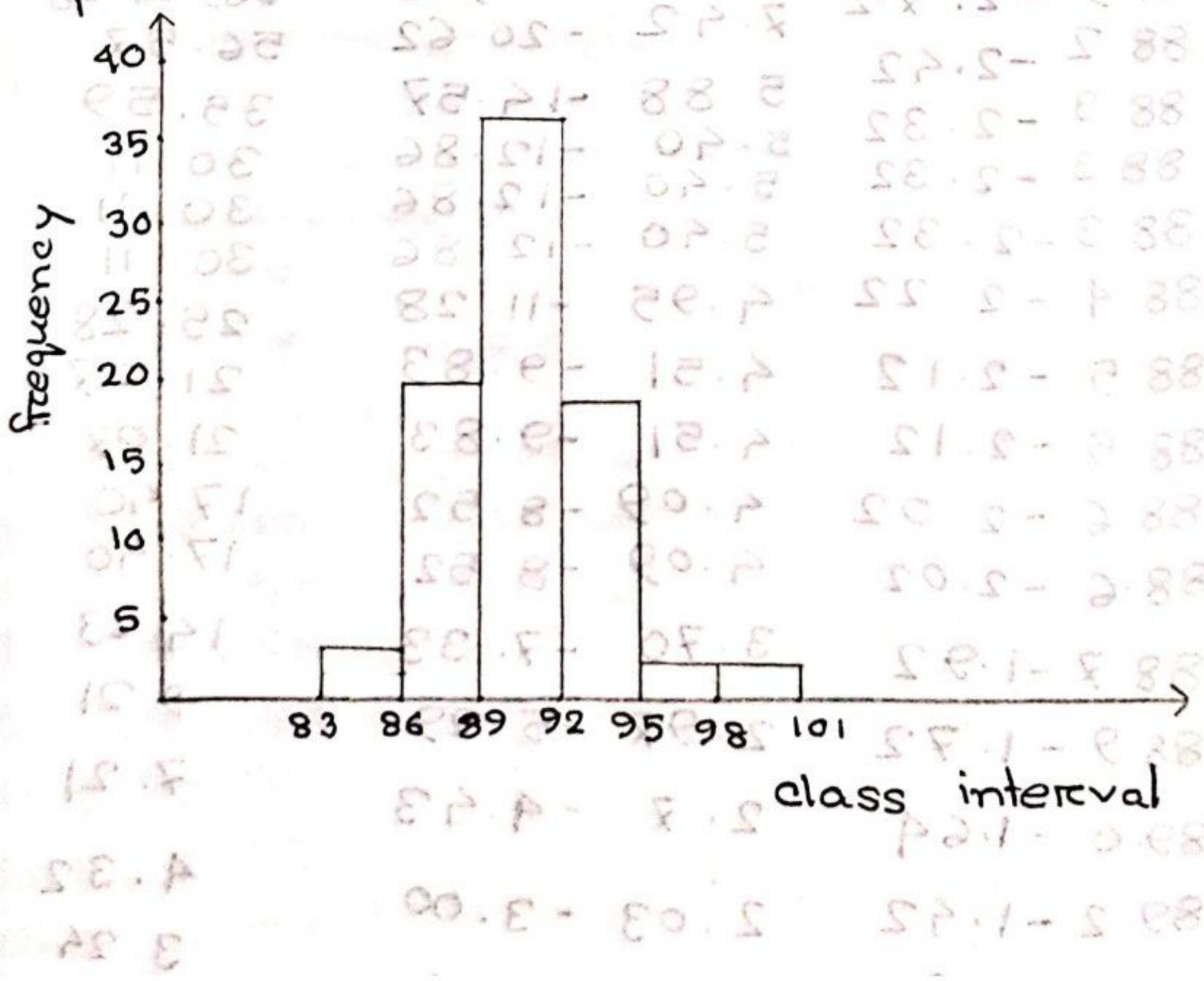
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Frequency distribution table:

class interval	Frequency
83 - 86	3
86 - 89	19
89 - 92	36
92 - 95	18
95 - 98	2
98 - 101	2
total	80

histogram:



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c) Sample mean, Sample varciance, Sample deviation: Mean $= \bar{x} = \sum_{i=1}^{n=80} \frac{x_i}{80} = \frac{7251.4}{80} = 90.669$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$(x_i - \bar{x})^3$	$(x_i - \bar{x})^4$
83.4	-7.34	52.42	-379.9	2751.4
84.3	-6.34	40.22	-255.142	1618.2
85.3	-5.34	28.54	-152.489	814.663
86.7	-3.92	15.40	-61.28	241.59
86.7	-3.92	15.40	-61.28	241.59
87.4	-3.24	10.51	-34.09	110.54
87.5	-3.22	10.39	-31.03	97.52
87.6	-3.02	9.14	-28.16	85.68
87.7	-2.92	8.55	-25.97	74.96
87.8	-2.82	7.98	-22.963	65.28
87.9	-2.72	7.42	-20.62	56.57
88.2	-2.42	5.88	-14.57	35.59
88.3	-2.32	5.40	-12.86	30.11
88.3	-2.32	5.40	-12.86	30.11
88.3	-2.32	5.40	-12.86	30.11
88.4	-2.22	4.95	-11.28	25.28
88.5	-2.12	4.51	-9.83	21.07
88.5	-2.12	4.51	-9.83	21.07
88.6	-2.02	4.09	-8.52	17.40
88.6	-2.02	4.09	-8.52	17.40
88.7	-1.92	3.70	-7.33	14.23
88.9	-1.72	2.97	-5.29	9.21
89.0	-1.64	2.7	-4.43	7.21
89.2	-1.42	2.03	-3.00	4.32
89.3	-1.32	1.75	-2.42	3.24

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x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$(x_i - \bar{x})^3$	$(x_i - \bar{x})^4$
89.3	-1.32	1.75	-2.42	3.24
89.6	-1.02	1.05	-1.13	1.18
89.7	-0.92	0.85	-0.79	0.73
89.8	-0.82	0.68	-.59	0.50
89.8	-0.82	0.68	-.59	0.50
89.9	-0.72	0.52	-.40	0.30
89.9	-0.72	0.52	-0.40	0.30
90.0	-0.62	0.39	-0.26	0.17
90.1	-0.52	0.28	-0.16	0.08
90.1	-0.52	0.28	-0.16	0.08
90.3	-0.32	0.10	-0.40	0.01
90.3	-0.32	0.10	-0.40	0.01
90.4	-0.22	0.50	-0.01	0.003
90.4	-0.22	0.50	-0.014	0.003
90.4	-0.22	0.50	-0.01	0.003
90.5	-0.12	0.02	-0.0028	0.0004
90.6	-0.02	0.0006	-7.67×10^{-5}	3.26×10^{-6}
90.7	0.07	0.0057	0.0002	1.09×10^{-5}
90.8	0.18	0.03	0.0039	0.0006
90.9	0.28	0.08	0.017	0.004
91.0	0.38	0.14	0.045	0.016
91.0	0.38	0.14	0.045	0.016
91.0	0.38	0.14	0.045	0.016
91.1	0.48	0.22	0.095	0.043
91.1	0.48	0.22	0.095	0.043
91.1	0.48	0.22	0.095	0.043
91.2	0.58	0.33	0.173	0.096
91.2	0.58	0.33	0.173	0.096
91.5	0.86	0.77	0.631	0.054
91.6	0.98	0.95	0.877	0.84

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x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$(x_i - \bar{x})^3$	$(x_i - \bar{x})^4$
91.6	0.98	0.95	0.877	0.84
91.8	1.18	1.38	1.55	1.79
91.8	1.18	1.38	1.55	1.79
92.2	1.57	2.48	3.77	5.88
92.2	1.57	2.48	3.77	5.88
92.2	1.57	2.48	3.77	5.88
92.3	1.68	2.80	4.55	7.54
92.6	1.98	3.90	7.5	14.68
92.7	2.08	4.31	8.71	17.92
92.7	2.08	4.31	8.71	17.92
92.7	2.08	4.31	8.71	17.92
93.0	2.38	5.64	13.10	30.28
93.2	2.58	6.63	16.72	42.78
93.3	2.68	7.16	18.76	49.87
93.3	2.68	7.16	18.76	49.87
93.4	2.78	7.70	20.96	57.81
93.8	3.08	9.46	28.58	87.39
94.2	3.58	12.78	45.023	160.169
94.2	3.58	12.78	45.023	160.15
95.6	4.98	24.75	121.84	604.02
96.81	5.48	29.98	162.55	887.11
98.8	8.18	66.84	542.84	4428.21
100.3	9.68	96.61	900.73	8698.79
94.4	3.78	14.25	53.05	199.34
94.7	4.08	16.61	66.81	271.04
	total	619.11	907.415	22259.1

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Sample variance,

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n=80} (x_i - \bar{x})^2$$

$$= \frac{619.11}{79}$$

$$= 7.84$$

Sample standard deviation

$$s = \sqrt{s^2}$$

$$= \sqrt{\frac{1}{n-1} \sum_{i=1}^{n=80} (x_i - \bar{x})^2}$$

$$= 2.7994$$

d) Median, Mode

here $n = 80$ which is even.

So, median, $Me = \frac{1}{2}\left[x_{\frac{80}{2}} + x_{\left(\frac{80}{2}+1\right)} \right]$

$$= \frac{1}{2}[x_{40} + x_{41}]$$

$$= \frac{1}{2}(90.4 + 90.5)$$

$$= 90.45$$

Mode:

Most frequent occurring in the sample is 3

So, modes are = 88.3, 90.4, 91.0, 91.1, 92.7

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e) Skewness and kurtosis measures

here, $S = 2.7994$

Skewness $\gamma_1 = \frac{n}{(n-1)(n-2)} \frac{\sum_{i=1}^{n}(x_i - \bar{x})^3}{s^3}$

$= \frac{80}{79 \times 78} \times \frac{907.42}{(2.7994)^3}$

$= 0.5370$

Kurtosis $\gamma_2 = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \frac{\sum_{i=1}^{n=80}(x_i - \bar{x})^4}{s^4} - 3$

$= \frac{80 \times 81}{79 \times 78 \times 77} \times \frac{22259.1}{(2.7994)^4} - 3$

$= 4.89 - 3$

$= 1.89$

As, $\gamma_1 > 0$, so the distribution is positively skewed.

As, $\gamma_2 > 0$, so the distribution is lepto kurtic.

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8.19)

a) Prove $\sum_{i=1}^{n} (x_i - \bar{x}) = 0$

Solve: We know.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i \quad \text{where } x_i = 1, 2 \cdots n$$

$$\sum_{i=1}^{n} x_i = n\bar{x} \quad \text{—— ①}$$

Now.

$$\sum_{i=1}^{n} (x_i - \bar{x})$$

$$= \sum_{i=1}^{n} x_i - \sum_{i=1}^{n} \bar{x}$$

$$= (x_1 - \bar{x}) + (x_2 - \bar{x}) + \cdots (x_n - \bar{x})$$

$$= \sum_{i=1}^{n} x_i - n\bar{x}$$

$$= \sum_{i=1}^{n} x_i - \sum_{i=1}^{n} x_i$$

$$= 0 \quad \text{(proved).}$$

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b) Prove $\sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$

Solution:

$$\sum_{i=1}^{n} (x_i - \bar{x})^2$$

$$= \sum_{i=1}^{n} \left(x_i - \frac{\sum_{i=1}^{n} x_i}{n}\right)^2$$

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$$= \frac{1}{n^2} \sum_{i=1}^{n} (nx_i - \sum_{i=1}^{n} x_i)^2$$

$$= \frac{1}{n^2} \sum_{i=1}^{n} \{n^2 x_i^2 - 2nx_i \sum_{i=1}^{n} x_i + (\sum_{i=1}^{n} x_i)^2\}$$

$$= \frac{1}{n^2} \left[\sum_{i=1}^{n} n^2 x_i^2 - 2n \sum_{i=1}^{n} nx_i \sum_{i=1}^{n} x_i + \sum_{i=1}^{n} (\sum_{i=1}^{n} x_i)^2\right]$$

$$= \frac{1}{n^2} \{n^2 \sum_{i=1}^{n} x_i^2 - 2n(\sum_{i=1}^{n} x_i)^2 + n(\sum_{i=1}^{n} x_i)^2\}$$

$$= \frac{1}{n^2} \{n^2 \sum_{i=1}^{n} x_i^2 - n^3 (\frac{\sum_{i=1}^{n} x_i}{n^2})^2\}$$

$$= \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$$

= R.H.S (proved)

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